

BIO503 Subjective Final Term Solved

1. What is molecule?

A group of atoms bonded together, representing the smallest fundamental unit of a chemical compound that can take part in a chemical reaction.

2. DNA microarray

A DNA microarray (also commonly known as DNA chip or biochip) is a collection of microscopic DNA spots attached to a solid surface.

Scientists use DNA microarrays to measure the expression levels of large numbers of genes simultaneously or to genotype multiple regions of a genome

3. Salting out:

It is a technique in which high concentrations of salts such as ammonium sulfate $[(\text{NH}_4)_2\text{SO}_4]$ are used to precipitate proteins.

Because each protein has a characteristic salting-out point, this technique removes many impurities.

0.8 M ammonium sulfate precipitates fibrinogen, a blood-clotting protein, whereas a concentration of 2.4 M is needed to precipitate serum albumin.

Salting out is also useful for concentrating dilute solutions of proteins, including active fractions obtained from other purification steps.

4. Animal sounds bioacoustics

- Sounds used by animals that fall within the scope of bioacoustics include a wide range of frequencies and media, and are often not "sound" in the narrow sense of the word (i.e. compression waves that propagate through air and are detectable by the human ear). Katydid crickets, for example, communicate by sounds with frequencies higher than 100 kHz.

5. First law of thermodynamics & act in living system?

The first law of thermodynamics is the application of the conservation of energy principle to heat and thermodynamic processes

- The first law makes use of the key concepts of internal energy, heat, and system work.

6. Active & passive electrolocation?

Active electrolocation	Passive electrolocation
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In active electrolocation, the animal senses its surrounding environment by generating electric fields and detecting distortions in these fields using electroreceptor organs. • This electric field is generated by means of a specialized electric organ consisting of modified muscle or nerves. Animals that use active electroreception include the weakly electric fish, which either generate small electrical pulses ("pulse-type") or produce a quasi-sinusoidal discharge from the electric organ (termed "wave-type"). • These fish create a potential which is usually smaller than one volt.	In passive electrolocation, the animal senses the weak bioelectric fields generated by other animals and uses it to locate them. • These electric fields are generated by all animals due to the activity of their nerves and muscles. A second source of electric fields in fish is the ion pumps associated with osmoregulation at the gill membrane. This field is modulated by the opening and closing of the mouth and gill slits. • Many fish that prey on electrogenic fish use the discharges of their prey to detect them. This has driven the prey to evolve more complex or higher frequency signals that are harder to detect.
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7. Function of cell membrane

Cell membrane have a crucial role in excitable cells such as neuronal and muscle tissues, allowing a rapid and coordinated depolarization in response to triggering voltage change.

8. Methods to reduce friction?

- **Friction can be reduced by making the surface smooth by polishing**
The friction is due to the roughness of surface. If we make the surfaces smooth by polishing ,then friction will be reduced. For example a slide in the park is polished to make its surface smooth and reduce friction.
- **Friction can be reduced by applying lubricants (like oil or grease) to the rubbing surface**
When oil or grease is applied between the moving parts of a machine ,a thin layer of oil or grease is formed between the two rubbing surfaces. This layer of oil separates the two rubbing surfaces a little bit due to which their interlocking is reduced to a large extent. This friction is reduced and movement becomes smooth.
- **Friction can be reduced by using wheels to move objects**
It is quite difficult to move a heavy suitcase by dragging it on the ground because the sliding friction between the heavy suitcase and the ground is very large. If this heavy suitcase is fitted with small wheels then it can be pulled very easily. Because when we attach wheels then sliding friction is converted into rolling friction.
- **Friction can be reduced by using ball bearing between the moving parts of machine**

Ball bearing is a device which consists of a ring of small metal balls. The small metal balls of a ball bearing can roll freely. They reduce friction by making the two surfaces (axle and wheel) to roll over each other.

9. What is diffusion? In animal and plant?

Diffusion is the movement of particles or substances from an area that is crowded and concentrated to an area that is not in order to reach an equilibrium.

Diffusion in Plants

Plants have no special organs for breathing. They have to rely on diffusion for their supplies of oxygen and carbon dioxide.

There are pores in the leaves and stems through which the gases diffuse. In daylight, CO₂ (for photosynthesis) will be diffusing in and O₂ will be diffusing out. In darkness, O₂ will diffuse in and CO₂ will diffuse out as a result of respiration.

The plant is respiring all the time, using up oxygen and producing carbon dioxide. In daylight, photosynthesis will use up the carbon dioxide as fast as it is produced by respiration, so there will be no outward diffusion of carbon dioxide. Similarly, the oxygen produced by photosynthesis is used up by respiration.

Diffusion in Animals

Diffusion is important to organisms because it is the process by which useful molecules enter the body cells and waste products are removed.

Digested food molecules (amino acids, glucose) move down a concentration gradient from the intestine to the blood.

Waste products such as carbon dioxide or urea travel by diffusion from body cells into the bloodstream.

Oxygen moves from high concentration (in the air sac) to a lower concentration (in the blood).

Carbon dioxide moves from high concentration (in the blood) to a lower concentration (in the air sac).

10. Define HPLC?

High-performance liquid chromatography (HPLC) formerly referred to as high-pressure liquid chromatography is a technique in analytical chemistry used to separate, identify, and quantify each component in a mixture. High performance liquid chromatography is a powerful tool in analysis, it yields high performance and high speed compared to traditional column chromatography because of the forcibly pumped mobile phase. HPLC is a chromatographic technique that can separate a mixture of compounds.

11. Define MRI (Magnetic Resonance Imaging)

A medical imaging technique which is used in radiology to form pictures of the anatomy and the physiological processes of the body in both health and disease.

MRI scanners use strong magnetic fields, electric field gradients, and radio waves to generate images of the organs in the body.

12. Electrophoresis

Electrophoresis separates mixtures of proteins based on charge, charge/mass ratio or size.

13. State Na⁺K⁺ATP function?

14. Translatory motion and its type?

In Translatory motion, the particle moves from one point in space to another. This motion may be along a straight line or along a curved path.

Following are 2 Types

Rectilinear Motion: Motion along a straight line is called rectilinear motion.

Curvilinear Motion: Motion along a curved path is called curvilinear motion.

15. State Animal use aquatic sound?

Some usually aquatic animals, such as sharks, have acute bioelectric sensors providing a sense known as electroreception, while migratory birds navigate in part by orienteering with respect to the Earth's magnetic field.

16. Biophysics

Biophysics is an interdisciplinary science that applies the approaches and methods of physics to study biological systems.

Biophysical research shares significant overlap with biochemistry, physical chemistry, nanotechnology, bioengineering, computational biology, biomechanics and systems biology.

17. Three applications of Ion Exchange Chromatography:

- Used in the analysis of amino acids. The amino acid “Auto analyzer” is based on ion exchange principle.
- Determine the base composition of nucleic acids. Chargaff used this technique for establishing the equivalence of Adenine and Thymine; Guanine and Cytosine.
- This is most effective method for water purification. Complete deionization of water (or) a non-electrolyte solution is performed by exchanging solute cations for hydrogen ions and solute anions for hydroxyl ions.

This is usually achieved by method is used for softening of drinking water. Proteins are also successfully separated by this technique.

18. Voltage-gated ion channels

Voltage-gated ion channels are a class of trans membrane proteins that form ion channels that are activated by changes in the electrical membrane potential near the channel.

19. Metabolic pathway

A sequence of chemical reactions, where the product of one reaction serves as a substrate for the next, is called a

metabolic pathway or biochemical pathway.

Most metabolic pathways take place in specific regions of the cell.

- Two types of metabolic pathway occur in a cell, Catabolic pathway and Anabolic pathway.

20. PROTEIN ISOLATION AND PURIFICATION

Protein isolation and purification achieved in several steps namely,

- Cell disruption
- Fractionation
- Crude sample preparation
- Purification

21. What is immune-blotting

Immunoblotting is a protein visualization technique applied in combination with affinity chromatography.

22. Photosynthesis

Most important transfer event, for life on our planet is the biological process called photosynthesis, which occurs in most plants, algae, and some bacteria.

Photosynthesis is the process wherein light energy from the sun is converted into chemical energy to make organic compounds, or food (mainly sugars), for the organism.

23. Plant bioacoustics

Measured sound emissions by plants as well as differential germination rates, growth rates and behavioral modifications in response to sound are well documented.

Plants detect neighbors by means other than well-established communicative signals including volatile chemicals, light detection, direct contact and root signaling.

Because sound waves travel efficiently through soil and can be produced with minimal energy expenditure, plants may use sound as a means for interpreting their environment and surroundings.

24. Define energy:

Energy is very important in our life, where we can't carry out different activities (mental or muscular)

without the produced energy from burning sugar inside our bodies , we can't cook our food without the

heat energy produced from the burning of natural gas.

There are various forms of energy like Chemical energy, Heat energy , Light energy , Electrical energy and Kinetic energy.

25. SPECT

SPECT is a nuclear imaging scan that integrates computed tomography (CT) and a radioactive tracer. The tracer is what allows doctors to see how blood flows to tissues and organs. Before the *SPECT* scan, a tracer is injected into your bloodstream.

26. Radioactive decay

The spontaneous transformation of an unstable atomic nucleus into a lighter one, in which radiation is released in the form of alpha particles, beta particles, gamma rays

27. Biophysical techniques

Biophysical techniques provide information about the electronic structure, size, shape, dynamics, polarity, and modes of interaction of biological molecules. Some of the most exciting techniques provide images of cells, subcellular structures, and even individual molecules.

28. direct effect ionizing radiation:

When cells are exposed to **ionizing radiation**, radiochemical damage can occur either by **direct action** or **indirect action**. **Direct action** occurs when alpha particles, beta particles or x-rays create ions which physically break one or both of the sugar phosphate backbones or break the base pairs of the DNA.

29. Reverse Phase HPLC:

The stationary phase is nonpolar (hydrophobic) in nature, while the mobile phase is a polar liquid, such as mixtures of water and methanol or acetonitrile. It works on the principle of hydrophobic interactions hence the more nonpolar the material is, the longer it will be retained.

30. Components Of Gel Filtration

The basis components of Gel filtration experiment are:

- i. Matrix
- ii. Chromatography column
- iii. Elution buffer

31. ELECTRODIALYSIS

Electro dialysis (ED) is used to transport salt ions from one solution through ion-exchange membranes to another solution under the influence of an applied electric potential difference.

32. Instruments of NMR

NMR Instrumentation

- 1. Sample holder :-** Glass tube with 8.5 cm long, 0.3 cm width
- 2. Permanent magnet :-** It provides homogeneous magnetic field at 60-100 MHz
- 3. Magnetic coils :-** These coils induce magnetic field when current flows through them.
- 4. Sweep generator :-** To produce the equal amount of magnetic field pass through the sample

5. Radio frequency (radio transmitter coil) :- A radio transmitter coil that produces a short powerful pulse of radio waves

6. Radio frequency (radio receiver coil) :- A radio receiver coil that detects Receiver radio frequencies

33. Explain 3rd law of thermodynamics? Taking water as examples?

The Third Law of Thermodynamics can be envisioned by taking water as an example.

- Water in gas form has molecules that can move around very freely. Water vapor has very high entropy (randomness).
- As the gas cools, it becomes liquid. The liquid water molecules can still move around, but not as freely. They have lost some entropy.
- When the water cools further, it becomes solid ice. The solid water molecules can no longer move freely, but can only vibrate within the ice crystals.
- The entropy is now very low. As the water is cooled more, closer and closer to absolute zero, the vibration of the molecules diminishes.
- If the solid water reached absolute zero, all molecular motion would stop completely. At this point, the water would have no entropy (randomness) at all.

34. Difference between intermolecular and intramolecular forces

Intramolecular forces are the forces that hold atoms together within a molecule. Intermolecular forces are forces that exist between molecules.

The four key intermolecular forces are as follows: Ionic bonds > Hydrogen

bonding > Van der Waals dipole-dipole interactions > Van der Waals dispersion forces.

Examples of intermolecular forces include the London dispersion force, dipole-dipole interaction, ion-dipole interaction, and van der Waals forces.

35. Ion-dipole force:

An ion-dipole force is an attractive force that results from the electrostatic attraction between an ion and a neutral molecule that has a dipole.

Most commonly found in solutions. Especially important for solutions of ionic compounds in polar liquids.

A positive ion (cation) attracts the partially negative end of a neutral polar molecule.

A negative ion (anion) attracts the partially positive end of a neutral polar molecule.

36. covalent bond

A **covalent bond**, also called a molecular **bond**, is a chemical **bond** that involves the sharing of electron pairs between atoms.

37. Detail note on Endocytosis:

Endocytosis is the process in which cells absorb molecules by engulfing them. The plasma membrane creates a small deformation inward, called an invagination, in which the substance to be transported is captured.

There are two main kinds of endocytosis:

Phagocytosis, or cellular eating, occurs when the dissolved materials enter the cell. The plasma membrane engulfs the solid material, forming a phagocytic vesicle.

Pinocytosis, or cellular drinking, occurs when the plasma membrane folds inward to form a channel allowing dissolved substances to enter the cell, as shown in Figure below. When the channel is closed, the liquid is encircled within a pinocytic vesicle.

38. Smectic & Nematic phase

In the nematic phase, only the long axes of the molecules are aligned, whereas in the smectic phase, the long axes of the molecules are parallel and the molecules are arranged in planes.

39. 2 major Forces of attractions:

There are two kinds of forces, or attractions, that operate in a molecule.

1. Intra-molecular.
2. Intermolecular.

40. Two major techniques of fractionation for fractionation of natural samples. 2 marks

Two main methods for fractionation of natural samples are as:

1. Bioassay-guided fractionation
2. Blood fractionation

41. Direct effects of ionizing radiation

There are two different types of damaging biological effects due to ionizing radiation on the cellular level.

1. Exposure can cause either cell death, or
2. It can change its genetic information through mutations, leaving the cell able to reproduce itself.

42. Which one have higher boiling point H₂O or N₂

The boiling point of H₂O is hundreds of degrees greater than the boiling point of N₂ because of stronger molecular interactions in H₂O (liq) than in N₂ (liq).

The forces between molecules in H₂O(liq) are greater than those in N₂ (liq).

43. Amino acids

Amino acids are the building blocks of proteins and are extremely important in biology. Twenty percent of the human body is made up of protein.

Protein plays a crucial role in almost all biological processes and amino acids are the building blocks of it.

The chain of amino acids then folds into a unique shape. That shape allows the enzyme to carry out specific chemical reactions.

44. Microsievert and millisievert

microsievert (μSv) is one millionth of a sievert ($1/1,000,000$)

millisievert (mSv) is one thousandth of a sievert ($1/1,000$)

45. Saxitoxin

Saxitoxin is a potent neurotoxin and the best-known paralytic shellfish toxin. Ingestion of saxitoxin by humans, usually by consumption of shellfish contaminated by toxic algal blooms, is responsible for the illness known as paralytic shellfish poisoning.

46. chiral phase

The thickness of the chiral phase is such that polarized light passing through it is rotated by 90° , which corresponds to the orientation of the right-hand polarizing filter.

So in this state, the light passes through and illuminates the pixel surface.

47. Functions of the cell membrane:

- To maintain the physical integrity of the cell – that is to mechanically enclose the contents of the cell, and also.
- To control the movement of particles e.g. ions or molecules, into and out of the cell.
- The cell membrane maintains the physical integrity of the cell.
- It's most obvious in the cases of animal cells (because they don't have cell walls) that the cell membrane holds the cell together by enclosing the cytoplasm and organelles within it.
- The cell membrane forms a barrier between the inside of the cell and the environment outside the cell – enclosing cytoplasm and any organelles within the cell, and enabling different chemical environments to exist on each side of the cell membrane.

48. Kinetic friction

Sliding friction occurs between objects as they slide against each other. It acts in the direction opposite to the direction of motion. It prevents the object from moving too fast. It is also called as kinetic friction.

49. Name three Amino acid?

histidine, isoleucine, leucine, lysine, methionine, phenylalanine, threonine, tryptophan, and valine.

50. Active electrolocation & Passive electrolocation

Active electrolocation

In active electrolocation, the animal senses its surrounding environment by generating electric fields and detecting distortions in these fields using electroreceptor organs.

This electric field is generated by means of a specialized electric organ consisting of modified muscle or nerves.

This field may be modulated so that its frequency and wave form are unique to the species and sometimes, the individual.

Passive electrolocation

In passive electrolocation, the animal senses the weak bioelectric fields generated by

other animals and uses it to locate them.

These electric fields are generated by all animals due to the activity of their nerves and muscles. A second source of electric fields in fish is the ion pumps associated with osmoregulation at the gill membrane. This field is modulated by the opening and closing of the mouth and gill slits.

Many fish that prey on electrogenic fish use the discharges of their prey to detect them. This has driven the prey to evolve more complex or higher frequency signals that are harder to detect.

51. Cofactors:

Cofactors are non-protein enzyme helpers. Cofactors may be inorganic (such as a metal in ionic form) or organic.

52. Coenzyme

An organic cofactor is called a coenzyme. Coenzymes include vitamins.

53. Active site

The active site is the region on the enzyme where the substrate binds.

54. Radiation protection

The protection of people from harmful effects of exposure to ionizing radiation, and the means for achieving this.

55. Fluid mosaic model

The fluid mosaic model describes the structure of a cell membrane. It indicates that the cell membrane is not solid. It is flexible and has a similar consistency to vegetable oil, so all the individual molecules are just floating in a fluid medium, and they are all capable of moving sideways within the cell membrane.

VU Biology Zone